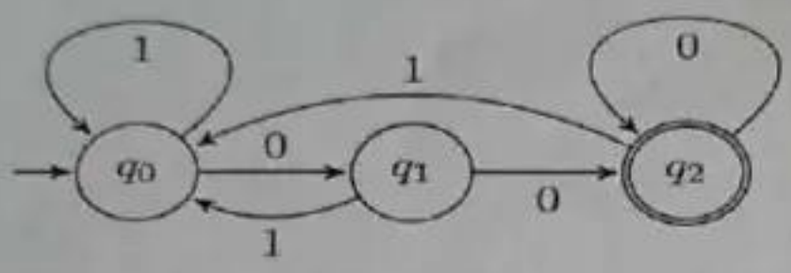
5th batch 6th semester

Department of Computer Science and Engineering
University of Barisal
3rd Year 2nd Semester Final Examination, 2020

Course Code: CSE-3203 | **Course Title: Theory of Computation**
Total time: 3.00 hours | **Total marks: 60**

(Note: Answer any five set of questions from the followings)

1. a) Construct a context-free grammar for the following DFA: 4



b) In compiler, ambiguous testing is a very important matter. Show that the grammar ({S}, {a, b}, R, S) with rules R = S → aS | aSbS | ϵ is ambiguous. 6

c) Does a push down automata have memory? Justify. 2

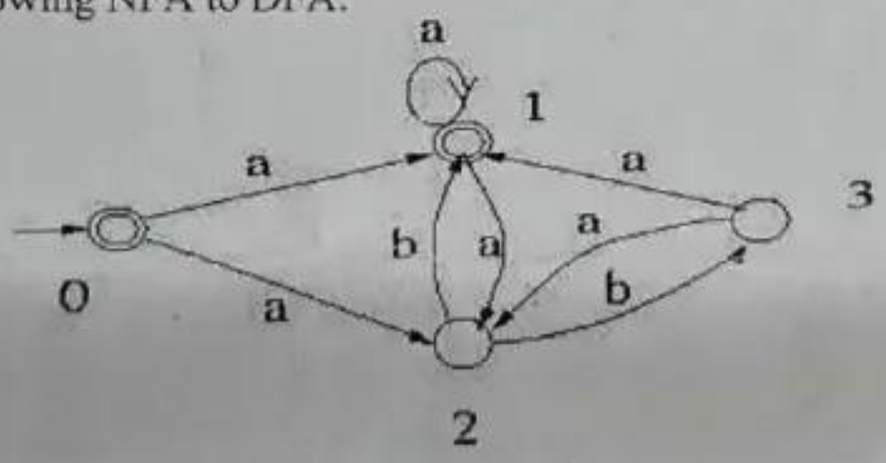
2. a) Why explicit ε-transitions in finite automata is important? 2

b) Build an ε-NFA for the following language: 6
L = {w | w is empty, or if non-empty will end in 01}

c) Convert ε-NFA to DFA based on Question 2(b) 4

3. a) Differentiate between Finite State and Turing Machines. 3

b) Convert the following NFA to DFA. 6



c) How a DFA processes strings? 3

3

4. a) Define push down automata with example. 3

b) Give pushdown automata that recognize the following languages: 5
(a) A = { w ∈ {0, 1}* | w contains at least three 1s }
(b) B = { w ∈ {0, 1}* | w = w^R and the length of w is odd }

c) Use the pumping lemma to prove that the language A = { $0^{2n} 1^{3n} 0^n$ | n ≥ 0 } is not context free. 4 6

5. a) Let, ∑ = {0, 1}. Construct a DFA for the following language:

L = {w | w is a binary string that has even number of 1s and even number of 0s}

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